

The Semantics of Thinking: Human Cognition as a Guide for LLM Optimisation

Differentiation Report

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Abstract

This report investigates the relationship between semantic representations in large language models and those observed in human cognition. The project combines methods from computational linguistics, cognitive neuroscience, and representation learning to compare model-derived signals with human neural and behavioural data. Initial experiments examine whether surprisal values from models such as LLaMA, GPT-2, Mistral, and BERT align with fMRI responses during naturalistic story listening. Further work uses representational similarity methods to compare the geometry of model and neural representations, and explores whether cognitively grounded similarity signals can be incorporated directly into training objectives through representational similarity regularisation. The overall aim is to better understand how language models represent meaning, and whether alignment with human semantic structure can lead to more interpretable and cognitively informed model optimisation.

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1 Introduction

Large language models (LLMs) have advanced rapidly over the past decade, evolving from relatively simple statistical systems into highly sophisticated neural architectures. Earlier approaches, such as n-gram models and recurrent neural networks (RNNs), were constrained in their ability to preserve information across longer contexts. The introduction of the Transformer architecture by Vaswani et al. [1] in 2017 marked a major turning point. By using self-attention, Transformers made it possible for models to consider all parts of a sentence simultaneously. This made language modelling far more scalable and laid the foundation for powerful systems such as BERT, ChatGPT, and LLaMA.

Despite this progress, LLMs still fall short of genuine language understanding. Although modern models can produce fluent and convincing text, their outputs are generated from statistical patterns in data rather than from any grounded understanding of meaning. This concern aligns with classic thought experiments in artificial intelligence, including Searle’s Chinese Room [2] and the Turing Test [3]. In both cases, the central question is whether an artificial system truly understands language or simply produces behaviour that gives that impression.

This study turns to the human brain as a way of beginning to address that gap. Decades of research in cognitive neuroscience suggest that the brain relies on predictive and semantic processes to integrate information across varying timescales and modalities during language tasks. Using neuroimaging methods such as functional magnetic resonance imaging (fMRI) and magnetoencephalography (MEG), researchers have examined how meaning is represented and processed across different brain regions. These methods offer valuable insights into real-time language comprehension by tracking the dynamic neural activity associated with reading, listening, and understanding language.

The aim of this project is to investigate whether the design of LLMs can be informed, and potentially improved, by what is known about how the human brain processes language. By comparing the internal representations of transformer-based LLMs with patterns of brain activity recorded during naturalistic language comprehension, we can begin to ask whether these models capture anything that is cognitively meaningful. If they do not, this comparison may also point towards ways in which their representations could be improved.

2 Background and Related Work

2.1 Semantic Representation in the Human Brain

Understanding how meaning is represented in the human brain has been a central question in cognitive neuroscience for decades. Early work in neurolinguistics identified specific cortical areas linked to language, especially Broca’s area in the left inferior frontal gyrus and Wernicke’s area in the posterior superior temporal gyrus. More recent research however, has shown that semantic processing is not limited to a small set of brain regions. Instead, it appears to arise from coordinated activity across a broader distributed network that includes temporal, frontal, and parietal cortices [4, 5].

Advances in neuroimaging have made it possible to study this process in much greater

detail. Functional magnetic resonance imaging (fMRI) offers relatively high spatial resolution, allowing researchers to identify which brain regions are involved in semantic processing during reading and listening. Magnetoencephalography (MEG), by contrast, provides millisecond-level temporal resolution, making it possible to track how semantic information develops over time as language is processed. Used together, these methods offer complementary perspectives on the neural basis of language comprehension.

One of the most widely studied neural markers of semantic processing is the N400 component, first described by Kutas and Hillyard [6]. The N400 is a negative-going brain response that peaks around 400 milliseconds after the presentation of a semantically unexpected word. Its amplitude varies depending on contextual fit and predictability, suggesting that the brain continuously forms expectations about upcoming linguistic input and revises its interpretation as each word is encountered [7]. This supports broader predictive accounts of language comprehension, in which meaning is actively constructed through the integration of context over time rather than retrieved in a purely passive, word-by-word fashion.

These findings are important for the present project because they suggest that semantic processing in the brain is both distributed and dynamic. Meaning is not stored in a single location or represented at a single moment. Instead, it emerges through evolving patterns of neural activity during comprehension. This makes the brain a useful point of comparison for language models, whose internal representations also develop across layers, context, and time.

2.2 Semantic Representation in Language Models

In parallel with advances in neuroscience, natural language processing has produced increasingly powerful language models. Early statistical methods, such as n-gram models, relied on local co-occurrence patterns and were limited in their ability to capture long-range dependencies. Neural approaches, including recurrent neural networks (RNNs) and long short-term memory networks (LSTMs), improved on this by maintaining hidden states across sequences. Even so, they still faced difficulties when scaling to very large corpora and longer contexts.

A major turning point came with the introduction of the Transformer architecture [1], which uses self-attention to model relationships between tokens across an entire sequence. This made possible the development of large-scale language models such as BERT [8], GPT-style models [9], and more recent open-weight systems such as LLaMA. These models learn vector representations of words and sequences from vast amounts of text and can perform a wide range of tasks with little or no task-specific supervision.

An important distinction can be drawn between static and contextual representations. Earlier embedding models such as Word2Vec assign each word a single vector regardless of context, whereas transformer-based models produce contextualised representations that change depending on the surrounding words. This matters for semantic modelling because many aspects of meaning are context-dependent.

Despite their strong empirical performance, language models remain difficult to interpret. Their representations are learned from statistical regularities in text rather than from direct grounding in perception or action, and it is therefore not always clear how closely they align with human conceptual organisation [10, 11]. This has led to growing interest in comparing language model representations with human behavioural and neural data, both to better

understand what these models encode and to assess whether they capture structure that is cognitively meaningful.

2.3 Brain-Language Model Comparisons

A growing body of research has sought to compare the internal representations of language models with patterns of neural activity recorded during language comprehension. These studies aim to determine whether artificial models and the human brain represent similar kinds of information, and, if they do, at what level and at what stage of processing this similarity appears.

An influential early example is the study by Wehbe et al. [12], who recorded MEG signals while participants read a chapter of *Harry Potter and the Sorcerer’s Stone (US)* one word at a time. A recurrent neural network language model processed the same text and generated contextual representations and prediction probabilities for each word. By comparing the neural signals with the model’s internal states, the authors found evidence that different components of the model aligned with different phases of neural processing. Neural activity before word onset was associated with contextual information, activity shortly after onset aligned with the model’s lexical representations, and later responses reflected the predictability of the incoming word. This study was important because it suggested that computational models and the brain may pass through partly similar stages during language comprehension.

Related research has examined the connection between language model predictions and neural markers of semantic expectancy. Frank et al. [13] showed that word surprisal, defined as the negative log probability of a word given its context, predicts the amplitude of the N400 response. This finding supports the idea that probabilistic language models capture aspects of the predictive processing carried out by the brain during comprehension.

More recent studies have extended these comparisons to modern transformer models. Large-scale evaluations have shown that internal representations from transformer-based language models can predict neural data recorded during naturalistic language processing, including fMRI responses elicited by stories and continuous speech [14]. These findings suggest that language models may capture representational structures that overlap, at least to some extent, with those used by the human brain.

This line of research provides an important foundation for the present project. It shows that comparisons between model signals and neural data are both feasible and informative, while also raising a further question: whether these comparisons can move beyond analysis and be used to guide model development itself.

2.4 Predictive Processing, Surprisal, and Neural Signals

A particularly important way of comparing language models with human cognition is through predictive processing. Both psycholinguistic and neuroscientific theories increasingly suggest that language comprehension involves the continuous generation of expectations about upcoming input. Rather than simply processing words one at a time as they appear, the brain seems to anticipate likely continuations and revise those expectations when incoming words are more or less surprising than predicted.

In computational terms, this predictive behaviour can be captured using surprisal. Surprisal measures how unexpected a word is in a given context and is typically defined as the negative log probability that a language model assigns to that word. Words with high surprisal are less expected, whereas words with low surprisal are more predictable given the preceding context. Because language models explicitly assign probabilities to possible continuations, surprisal provides a natural link between model-based predictions and human language processing.

Surprisal has been connected to several behavioural and neural measures of processing difficulty. In psycholinguistics, words with higher surprisal are often associated with longer reading times and greater processing effort. In neuroscience, surprisal has also been shown to correlate with neural responses linked to semantic integration and expectancy violation, including the N400 [13]. This makes surprisal a useful bridge between probabilistic language models and cognitive theories of language comprehension.

For the present project, surprisal is particularly relevant because it offers a concrete, model-derived signal that can be aligned with neural recordings collected during naturalistic story listening. It therefore provides an important first step in evaluating whether large language models capture predictive structure that is cognitively meaningful.

2.5 Representational Similarity Analysis

While surprisal focuses on prediction, another important way to compare brains and language models is by examining their internal representational structure. Representational Similarity Analysis (RSA), introduced by Kriegeskorte et al. [15], provides a general framework for this kind of comparison.

Rather than comparing raw representations directly, RSA compares the similarity structure that those representations produce across a set of stimuli. In practice, this involves constructing representational dissimilarity matrices (RDMs), where each cell captures the degree of similarity or dissimilarity between a pair of stimuli. For neural data, an RDM might be based on distances between patterns of brain activity elicited by different words or sentences. For a language model, an RDM can be built from distances between embedding vectors or hidden-state representations.

The strength of RSA is that it makes it possible to compare very different systems in a meaningful way. Brain activity, behavioural judgements, and model embeddings may differ in format and dimensionality, but each can be reduced to a shared relational structure. When two systems produce similar RDMs, this suggests that they encode the same stimuli in comparable ways, even if the underlying representations themselves are quite different.

RSA has become a standard method in cognitive neuroscience and has increasingly been applied in computational linguistics to compare language model representations with both behavioural and neural data. It is particularly useful for the present project because it offers a principled way to compare representational geometry across brains and models, while also providing the conceptual basis for the regularisation methods explored later in this research.

2.6 Representational Similarity Regularisation

Although similarity-based methods such as RSA are now widely used for analysis, relatively little research has examined whether similarity structure can be incorporated directly into the training of language models. Building on ideas developed by Ormerod [16], Representational Similarity Regularisation (RSR) extends the logic of RSA by turning similarity alignment into an optimisation objective.

The central idea is straightforward. Rather than training a model only to predict tokens or reconstruct inputs, it is possible to introduce an additional objective that encourages the model’s learned representations to reflect similarity structure observed in human data. These target similarity structures may be derived from behavioural judgements, neural recordings, or other cognitively meaningful sources. In this way, similarity is not simply measured after training but is actively used to shape representational geometry during learning, providing a richer and more semantically grounded signal than can be obtained from text alone.

This is particularly relevant to the present project because it offers a way to move from analysis to intervention. If neural or behavioural similarity structure captures aspects of human semantic organisation, then incorporating that structure into training may encourage language models to develop representations that are not only computationally useful but also more closely aligned with human cognition.

The later stages of this research therefore investigate whether representational similarity can be used not only to compare models with human cognition, but also to guide optimisation directly. This provides a natural extension of earlier work on brain-model comparison and opens up the possibility of training models whose internal semantic structure more closely reflects human language processing. Framing and implementing RSR in this way as a practical fine-tuning approach for modern language models represents a novel contribution to this area of research.

2.7 Behavioural Similarity Benchmarks and Semantic Evaluation

In addition to neural data, behavioural similarity judgements provide an important source of evidence about how humans organise meaning. Datasets such as MEN [17], SimVerb-3500 [18], THINGS [19], and SimLex-999 [20] contain human ratings of semantic similarity between pairs of words or concepts. These benchmarks offer a more direct measure of conceptual organisation than distributional statistics alone, because they reflect how people perceive semantic relationships between items.

Such datasets are valuable for two main reasons. First, they provide an intrinsic way of evaluating the semantic representations learned by language models. If a model assigns high similarity to concept pairs that humans also judge to be similar, this suggests that its embedding geometry captures meaningful aspects of semantic structure. Secondly, behavioural similarity data can also be used as supervision for training objectives such as RSR, allowing human semantic judgements to directly shape model optimisation.

This is especially important for the present project. While neural data provides a rich and cognitively grounded signal, behavioural similarity benchmarks offer a complementary and more scalable source of human semantic information. Taken together, these resources make it possible to examine whether cognitively informed constraints can improve language

model representations, both in terms of alignment with human judgements and in terms of broader semantic organisation.

Further details of these datasets are provided in Appendix A.

3 Research Project

The overall aim of this research is to investigate and improve the relationship between semantic representations in large language models and those observed in human cognition. In recent years, transformer-based language models have achieved impressive performance across a wide range of natural language processing tasks. Even so, it remains unclear to what extent their internal representations reflect the kinds of semantic structure used by the human brain during language comprehension.

This project examines that relationship from two complementary angles. First, it investigates the extent to which signals derived from language models correspond to neural responses recorded during naturalistic language processing. Second, it considers whether insights from these comparisons can be used to improve language model representations. More specifically, the research explores whether cognitively grounded constraints, derived from neural or behavioural data, can be incorporated into training objectives to produce models with semantic representations that are more closely aligned with human cognition.

The work therefore moves from analysis towards intervention. It begins by comparing model-derived signals with brain activity and then asks whether representational alignment can be used to guide model optimisation directly.

The project is organised around the following research questions:

- **RQ1:** To what extent do predictive signals derived from large language models correspond to neural responses observed during naturalistic language comprehension?
- **RQ2:** How closely do the internal representational structures of language models align with human semantic similarity judgements?
- **RQ3:** Can representational similarity signals derived from human cognition be incorporated into the training of language models to improve their semantic representations?

3.1 Methodology

To address these research questions, the project combines methods from computational linguistics, cognitive neuroscience, and representation learning through several complementary experimental approaches. Each of the following components corresponds directly to one of the research questions outlined above.

First, predictive signals derived from language models are compared with neural recordings collected during naturalistic story listening, addressing RQ1. Using datasets such as the Narratives fMRI corpus [21], word-level surprisal values are computed from multiple language models and aligned with temporally aggregated brain activity. These signals are then convolved with a hemodynamic response function to approximate expected BOLD responses and correlated with neural time series extracted from language-related cortical regions.

Second, representational similarity methods are used to compare the geometry of semantic representations across models and human data, addressing RQ2. Representational similarity analysis (RSA) makes it possible to compare neural responses, behavioural similarity judgements, and language model embeddings by focusing on pairwise similarity structure rather than raw representations.

Finally, building on these analytical approaches, the project explores whether similarity structure derived from human cognition can be incorporated directly into model optimisation, addressing RQ3. Representational Similarity Regularisation (RSR) introduces an additional objective that encourages models to align their representational geometry with human similarity judgements. These experiments examine whether incorporating such constraints during training can improve semantic alignment while preserving the broader capabilities of modern language models.

3.2 Experiment Progress

3.2.1 Experiment 1: Brain-Model Alignment Using the Narratives Dataset

Building on earlier studies such as Ormerod et al. [16], my initial experimental work investigated how predictive signals derived from large language models relate to human brain activity during naturalistic language comprehension. Using the Narratives dataset [21], I analysed averaged fMRI recordings from 80 participants who listened to the story *Pieman* while undergoing scanning. The transcript was aligned at the word level using forced-alignment timings, and surprisal values were computed from several autoregressive and masked language models, including LLaMA-3 (3B), GPT-2, Mistral, and BERT. Here, surprisal refers to how unexpected a word is given the model’s prior context. It was first calculated at the token level and then aggregated into word-level values for alignment with the transcript.

Figure 1 illustrates this token-to-word aggregation process for the LLaMA pipeline. Because many words are split into multiple subword tokens, surprisal was computed separately for each token and then recombined into a single word-level score. This step was important for maintaining compatibility between the model output and the forced-alignment timestamps used for the neural data. It also served as a useful sanity check, confirming that the pipeline handled tokenisation correctly before any comparison with brain activity was carried out.

	original_word	token	token_id	surprisal	top_predicted	actual_rank	start_ts	end_ts	word_index	word_surprisal
0	I	I	40	NaN	None	NaN	15.089999	15.169999	0	NaN
1	began	b	65	12.217824	Tags	299.0	15.170000	15.510000	1	12.217824
2	began	egan	16133	10.329922	iza	184.0	15.170000	15.510000	1	12.217824
3	my	my	2465	6.441408	to	12.0	15.509999	15.699999	2	6.441408
4	illustrious	ill	484	13.356200	first	1027.0	15.710000	16.310000	3	13.356200
5	illustrious	ust	592	4.607044	umin	4.0	15.710000	16.310000	3	13.356200
6	illustrious	rious	84718	1.786462	rious	1.0	15.710000	16.310000	3	13.356200
7	career	career	95637	1.426493	career	1.0	16.330000	16.940000	4	1.426493
8	in	in	258	2.523307	in	1.0	17.080000	17.210000	5	2.523307
9	journalism	journal	43618	7.867196	the	24.0	17.209999	17.929999	6	7.867196
10	journalism	ism	2191	0.080791	ism	1.0	17.209999	17.929999	6	7.867196
11	in	in	258	3.057274	with	2.0	18.469999	18.589999	7	3.057274
12	the	the	1820	3.527840	199	4.0	18.590000	18.680000	8	3.527840
13	Bronx	Bron	42059	12.206553	early	399.0	18.680000	19.240000	9	12.206553
14	Bronx	x	87	0.264341	x	1.0	18.680000	19.240000	9	12.206553
15	where	where	2940	6.813353	,	16.0	19.240000	19.390000	10	6.813353
16	I	I	40	0.927749	I	1.0	19.390000	19.500000	11	0.927749
17	toiled	to	998	9.388789	was	73.0	19.500000	20.090000	12	9.388789
18	toiled	iled	2230	5.739785	ok	7.0	19.500000	20.090000	12	9.388789
19	as	as	300	2.399326	at	2.0	20.360000	20.490000	13	2.399326

Figure 1: Example of token-level surprisal output and aggregation to aligned word-level surprisal values for the *Pieman* transcript. Multi-token words were recombined so that model-derived surprisal could be matched to the word timings used in the fMRI analysis.

Each model’s word-level surprisal sequence was then binned into TR-sized intervals and convolved with a hemodynamic response function (HRF) to approximate the expected BOLD response. These convolved surprisal signals were compared with fMRI time series extracted from anatomically defined regions using the Harvard-Oxford cortical atlas. To make the spatial distribution of the results easier to interpret, the correlation maps were visualised separately for the left and right hemispheres.

Figures 2 and 3 show representative voxel-wise correlation maps for the LLaMA surprisal signal. Positive correlations were observed in several plausible language-related and associative regions, while negative correlations appeared in a smaller number of areas. Although these maps should be interpreted as exploratory rather than conclusive, they indicate that the model-derived predictive signal was not distributed randomly across the brain. Instead, the correspondence was concentrated in cortical regions broadly consistent with previous research on narrative comprehension and predictive language processing.

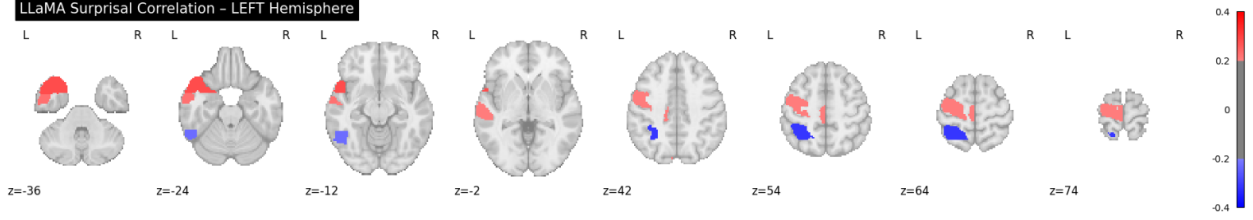


Figure 2: Voxel-wise correlation map for the left hemisphere showing the relationship between LLaMA-derived surprisal and fMRI activity during listening to *Pieman*. Warm colours indicate positive correlations and cool colours indicate negative correlations.

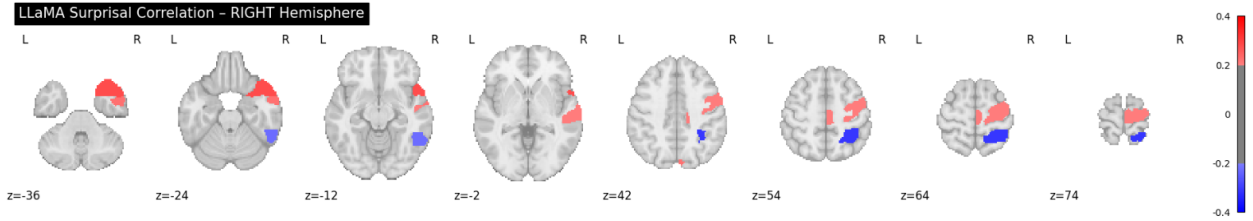


Figure 3: Voxel-wise correlation map for the right hemisphere showing the relationship between LLaMA-derived surprisal and fMRI activity during listening to *Pieman*. The bilateral pattern suggests that model-based predictive signals relate to activity in distributed regions rather than a single localised site.

To summarise the strongest effects more clearly, regions of interest were ranked according to their correlation with the convolved surprisal signal. Table 1 presents the top ten regions for the LLaMA model. The highest-ranking areas included the temporal pole, inferior temporal gyrus, middle temporal gyrus, superior temporal gyrus, and cuneal cortex, along with additional effects in precentral and supplementary motor regions. Several of these temporal areas are closely linked to semantic and narrative processing, which provides a useful sanity check on the plausibility of the findings. At the same time, the involvement of some visual and motor-related regions suggests that the signal captured by surprisal is not purely semantic, but may also reflect broader aspects of attention, imagery, or event processing during story comprehension.

Rank	Region of Interest	r	p -value
1	Temporal Pole	0.278	8.588×10^{-6}
2	Inferior Temporal Gyrus (anterior)	0.224	3.696×10^{-4}
3	Middle Temporal Gyrus (anterior)	0.221	4.663×10^{-4}
4	Cuneal Cortex	0.220	4.758×10^{-4}
5	Superior Temporal Gyrus (posterior)	0.206	1.112×10^{-3}
6	Precentral Gyrus	0.204	1.224×10^{-3}
7	Juxtapositional Lobule Cortex (SMA)	0.195	2.054×10^{-3}
8	Middle Temporal Gyrus (temporo-occipital)	0.183	3.773×10^{-3}
9	Intracalcarine Cortex	0.172	6.524×10^{-3}
10	Background/ Non-labelled	0.169	7.522×10^{-3}

Table 1: Top ten regions of interest ranked by correlation with the LLaMA surprisal signal.

Overall, the experiment showed that model-derived surprisal signals could be aligned in a meaningful way with human neural responses during naturalistic story listening. Although the correlations were modest in magnitude, they were still interpretable in anatomical terms, and the spatial distribution of the strongest effects was broadly consistent with earlier work linking predictive language processing to temporal, frontal, and associative cortical regions. In the wider context of the project, this experiment played an exploratory role. It helped establish the end-to-end processing pipeline, provided practical experience with large-scale neuroimaging datasets, and motivated the later shift from prediction-based comparisons towards representational similarity analysis and cognitively informed model optimisation.

3.2.2 Experiment 2: Representational Similarity Analysis on Naturalistic Story Stimuli

The second experiment extended the earlier brain-model alignment work by shifting the focus from word-level predictive signals to representational structure. Rather than correlating surprisal directly with neural activity, this experiment used Representational Similarity Analysis (RSA) to compare the geometry of language model representations with the geometry of fMRI responses recorded during naturalistic story comprehension.

Using stimuli from the Narratives framework, word-level forced-alignment files (`align.csv`) were used to reconstruct the story transcript and divide it into either fixed time windows or sentence-like chunks - dependant on user-choice. For each chunk, contextual embeddings were extracted from transformer language models, including GPT-2 and BERT. Pairwise dissimilarities between chunk representations were then computed to form a model representational dissimilarity matrix (RDM). Figure 4 shows an example of the chunked *Pieman* transcript used in the analysis. In this representative run, the story was divided into 67 chunks, producing a sequence of semantically meaningful narrative units rather than isolated words. This was important because RSA relies on comparing patterns across stimuli, and continuous story chunks provide a more ecologically valid basis for comparison than decontextualised sentence pairs or word lists.

Story Transcript

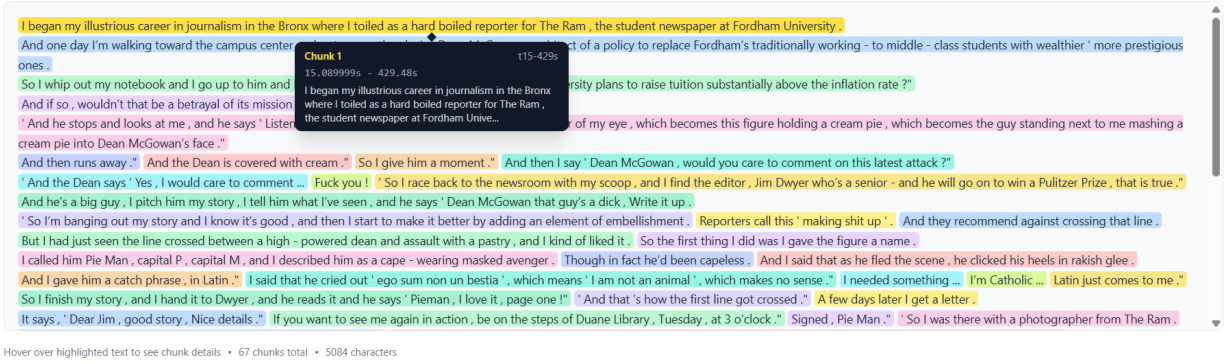


Figure 4: Chunked transcript view for the *Pieman* story used in Experiment 2. Forced-alignment information was used to reconstruct the narrative and divide it into 67 chunks, each of which served as a unit for embedding extraction and RSA comparison.

On the neural side, subject-specific fMRI data was loaded either from precomputed matrices or from BIDS-formatted Narratives data and then parcellated into cortical regions. For each parcel, a corresponding brain RDM was computed across the same sequence of text chunks. RSA was then carried out by correlating the upper triangles of the model and brain RDMs using Spearman correlation, producing a parcel-wise measure of brain-model representational alignment. The parcels were subsequently ranked by RSA score, and summary statistics were calculated across the full cortical set.

Figure 5 summarises the resulting parcel-wise distribution for a representative run on *Pieman* using subject **sub-001**. Across 986 Schaefer parcels (i.e., predefined cortical regions derived from a functional brain atlas), the overall mean RSA score was 0.129, while the highest-scoring parcel reached 0.292. Importantly, the alignment was not spatially uniform. The strongest parcels were concentrated in higher-order associative networks, including control, default-mode, orbitofrontal/limbic, and temporo-parietal regions, whereas the lowest-scoring parcels were more often linked to sensory or posterior systems. This pattern provides a useful sanity check on the analysis, since semantic and narrative representations would be expected to align more strongly with integrative cortical systems than with regions less directly involved in language comprehension.

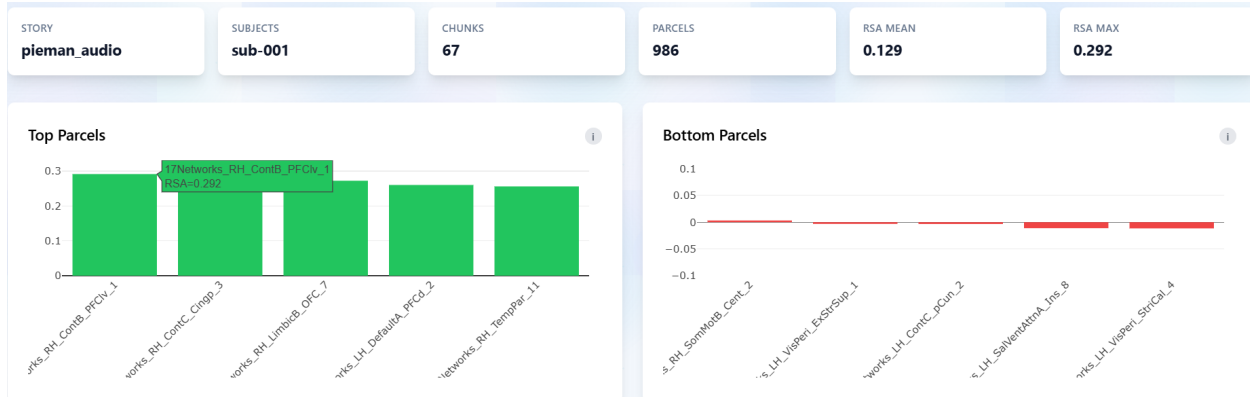


Figure 5: Parcel-wise RSA summary for a representative *Pieman* run. The dashboard reports 67 chunks, 986 parcels, an overall mean RSA of 0.129, and a maximum parcel-wise RSA of 0.292. Parcels refer to anatomically defined cortical regions from the Schaefer parcellation, which divides the cortex into functionally meaningful areas. The strongest parcels were found in cognitively plausible higher-order associative networks rather than uniformly across cortex.

Inspection of the representational dissimilarity matrices provided further support for this interpretation. The model RDM showed clear fine-grained structure across story chunks, with repeated banding and local block-like patterns suggesting that the embedding space was sensitive to narrative organisation and contextual similarity. The best-matching brain parcel produced a smoother and coarser RDM, which is what one would expect given the temporal blurring and noise associated with fMRI. Even so, it preserved enough relational structure to yield a positive RSA correlation. In the example shown in Figure 6, the best parcel achieved an RSA value of 0.174 when its RDM was compared with the model RDM. Although the raw distance scales differed between the two matrices, RSA is concerned with the relative ordering of pairwise dissimilarities rather than exact equality of values. The key point, therefore, is that the neural and model representations preserved partly similar similarity structure across narrative segments.

The observed correspondence, despite differences in scale and smoothness, suggests that transformer-based representations capture aspects of the underlying semantic structure that are also reflected in neural activity during naturalistic comprehension.

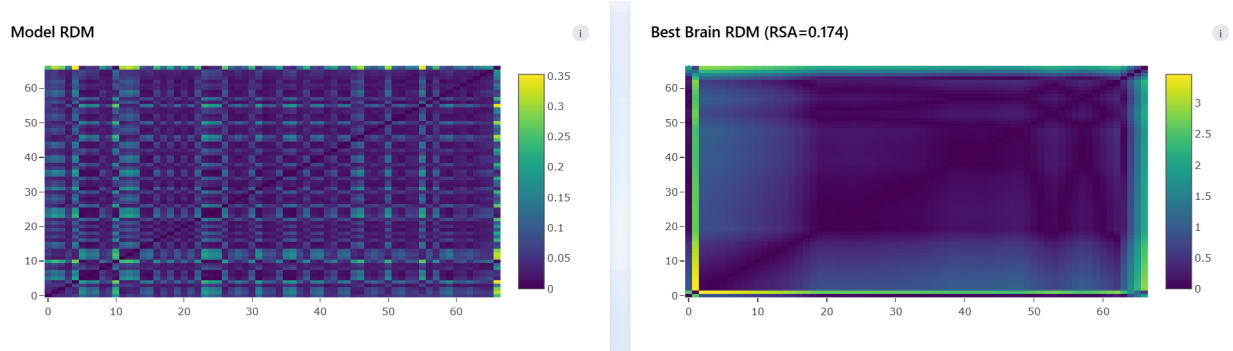


Figure 6: Model and best-matching brain RDMs for a representative RSA run using BERT embeddings. Both matrices are indexed by narrative chunks (aligned to fMRI timepoints, or TRs) along the x- and y-axes. The model RDM shows structured variation across narrative segments, while the brain RDM is smoother but retains partially similar relational structure, yielding a parcel-wise RSA of 0.174.

In this example, contextual embeddings were extracted using BERT, and both the model and brain RDMs are organised over the same sequence of narrative chunks aligned to fMRI timepoints (TRs).

In addition to the baseline RSA analysis, the pipeline supported several exploratory variants, including sentence-based chunking, temporal shifting to account for haemodynamic lag, HRF-inspired signal transformations, and subject-to-subject noise modelling as an approximate reliability ceiling. The experiment was implemented as an interactive analysis pipeline with a Python backend and a lightweight web interface, which made it possible to vary the model, story, subject, and signal settings systematically.

This experiment served as a methodological bridge between the first and third stages of the project. Where Experiment 1 focused on predictive signals such as surprisal, Experiment 2 examined the broader similarity structure of model and brain representations. The results were quantitatively meaningful and interpretable. Alignment was positive overall, strongest in plausible higher-order cortical systems, and visible in the partial correspondence between model and brain RDM geometry. Experiment 2 therefore provided both the conceptual and technical foundation for the later work on representational similarity regularisation.

3.2.3 Experiment 3: Representational Similarity Regularisation for Language Models

Building on the earlier exploratory work comparing language model predictions with neural signals, the next stage of the project examined whether representational alignment with human semantic structure could be incorporated directly into language model training. While many previous studies have used representational similarity methods to analyse model behaviour, relatively little research has investigated whether such signals can be used as an explicit training objective. This experiment therefore explored a proof-of-concept method for incorporating human semantic similarity information directly into model optimisation through an approach termed *Representational Similarity Regularisation* (RSR).

The main motivation for this experiment comes from the fact that standard language model training objectives are entirely distributional. Models such as Word2Vec, BERT, and GPT-style architectures learn representations by predicting words from surrounding context. Although this process captures a large amount of statistical structure from text, it does not explicitly include human knowledge about conceptual similarity. As a result, distributional embeddings often reflect relatedness rather than genuine semantic similarity. For example, words such as *coffee* and *cup* may appear highly similar in embedding space because they frequently co-occur, even though they belong to different conceptual categories.

RSR addresses this limitation by introducing an additional learning signal derived from human similarity judgements. Instead of learning semantic structure only from distributional statistics, the model is encouraged to organise its representational geometry in a way that better reflects how humans judge similarity between concepts. In practical terms, this was implemented by sampling pairs of words with associated human similarity scores and encouraging the model’s embedding space to preserve the same relative ordering of those similarities. Predicted similarity between two words was computed using cosine similarity between their embedding vectors, and the optimisation objective was formulated as a differentiable approximation to Spearman correlation. This meant that training focused on preserving the rank structure of human judgements rather than matching raw scores directly.

More formally, for a batch of word pairs, the model produces a set of predicted similarity scores based on embedding cosine similarity. These predicted scores are then compared with the corresponding human similarity scores, and a loss is computed that penalises mismatches in their relative ordering. The resulting representational similarity loss can be viewed as encouraging the model to produce an embedding space in which pairs judged as more similar by humans are also closer together than less similar pairs. In this sense, the objective operates over the geometry of the embedding space rather than over individual predictions.

Operationally, RSR is applied in an interleaved fashion during training. At each similarity step, a mini-batch of word pairs is sampled from the similarity dataset, embeddings are extracted for each item, and all pairwise similarities within the batch are computed. These are then compared to the corresponding human similarity structure to produce a single scalar loss. This loss is backpropagated through the model, updating parameters so as to better align the embedding geometry with human judgements. These similarity-based updates are alternated with standard language modelling updates, allowing the model to retain its distributional learning signal while gradually incorporating cognitively grounded structure.

The similarity supervision used in these experiments was drawn from three widely used behavioural resources: MEN, SimVerb-3500, and the THINGS similarity dataset. These datasets capture human similarity judgements across a range of lexical categories, including nouns, verbs, and concrete object concepts. Each dataset was normalised independently before being combined into a single pool of similarity-labelled word pairs. During training, mini-batches of pairs were sampled from this pool and used to compute the similarity regularisation objective.

To examine how this objective interacted with different model architectures, experiments were carried out across three model classes representing different stages in the development of language modelling. The first was the Word2Vec skip-gram model, which learns static word embeddings directly from distributional statistics. The second was BERT, a bidirectional transformer that produces contextualised representations through masked language

modelling. The third was GPT-2, an autoregressive transformer trained using next-token prediction. This range of architectures made it possible to test whether similarity-based supervision behaves differently depending on the model’s existing representational structure.

For the Word2Vec experiments, models were trained from scratch using the skip-gram objective with negative sampling on a large English Wikipedia corpus. Similarity supervision was incorporated through this interleaved training schedule, in which distributional updates and similarity-based updates alternated during optimisation. Every k -th training step optimised the similarity objective, while the remaining steps optimised the standard skip-gram objective. As shown in Figure 7, this interleaved RSR strategy produced a clear and highly consistent improvement across all random seeds. Vanilla Word2Vec models achieved Spearman correlations on SimLex-999 of roughly 0.26-0.29, whereas the RSR-enhanced models reached values around 0.40-0.42. The effect was therefore not only positive but also substantial, suggesting that similarity supervision can meaningfully reshape static embedding geometry when introduced during training from scratch.

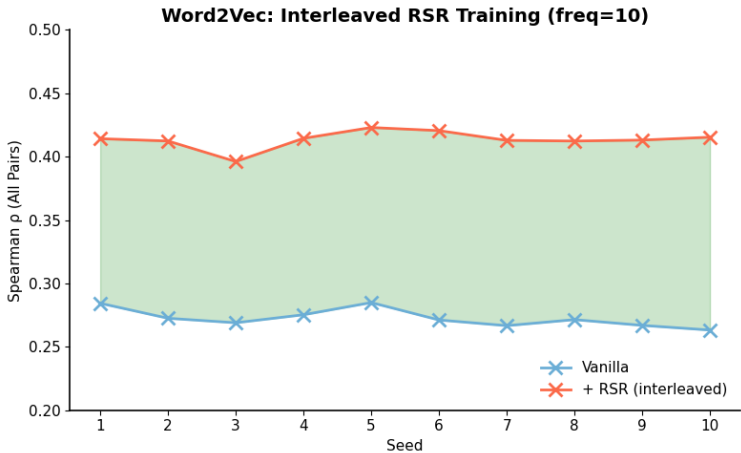


Figure 7: Word2Vec results under interleaved Representational Similarity Regularisation. Across all ten random seeds, the RSR-trained models outperform the vanilla baseline on SimLex-999, indicating that human similarity supervision consistently improves static embedding quality.

For the transformer-based models, the training procedure differed because the models had already been pre-trained on large corpora. Rather than training them from scratch, BERT and GPT-2 were fine-tuned using the similarity objective. A small projection head was added to map hidden representations into a lower-dimensional embedding space suited to similarity comparison. During fine-tuning, most transformer layers were frozen, allowing the projection head and final layers to adjust the representational geometry while preserving the models’ broader linguistic capabilities.

Model performance was evaluated using the SimLex-999 benchmark, which is designed to measure genuine semantic similarity rather than simple distributional relatedness. Performance was assessed using Spearman correlation between predicted similarities and human similarity scores. Figure 8 shows that RSR improved performance for both BERT and GPT-2 across all random seeds. For BERT, baseline performance was approximately 0.14-0.17,

while RSR increased this to around 0.27-0.30. For GPT-2, baseline performance was lower, at roughly 0.09-0.10, but RSR raised it to around 0.21-0.25. Although these improvements were smaller in absolute terms than those observed for Word2Vec, they were nevertheless robust and highly consistent across runs.

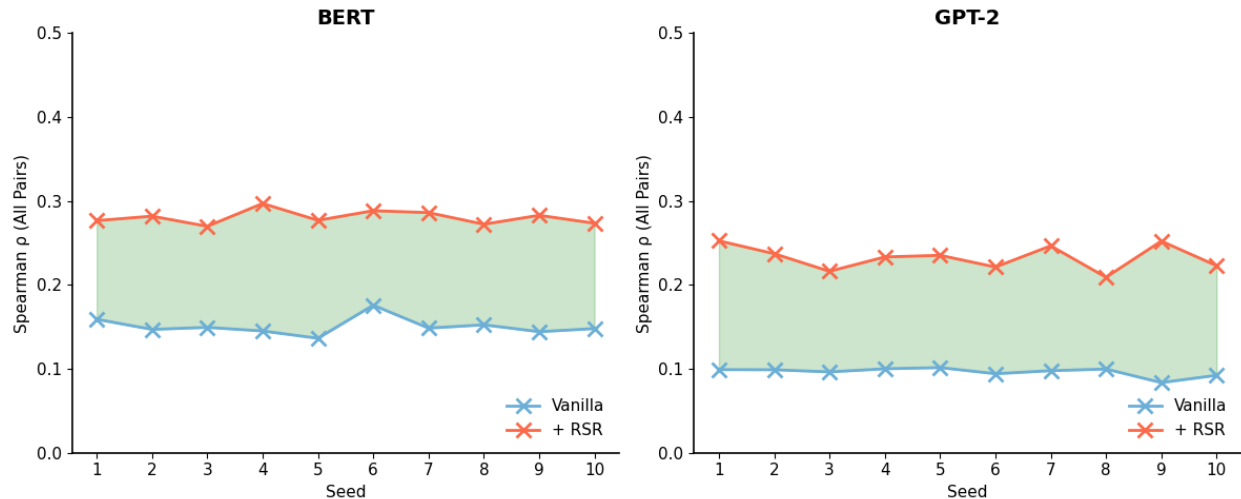


Figure 8: Transformer results for BERT and GPT-2 with and without Representational Similarity Regularisation. In both architectures, RSR improves SimLex-999 performance across all seeds, indicating that similarity-based fine-tuning consistently increases alignment with human semantic judgements.

The architecture-specific pattern of results is especially informative. For Word2Vec, the large improvement suggests that direct similarity supervision can substantially reorganise a static embedding space during training. For BERT and GPT-2, the gains show that cognitively grounded similarity information can also be introduced into contextual transformer representations, even when most of the underlying model remains frozen. At the same time, the improvements for the transformer models are more moderate, which is consistent with the view that these models already encode relatively rich semantic structure and may therefore be less dramatically altered by an additional supervision signal.

Taken together, the results suggest that representational similarity regularisation offers a viable way of incorporating human semantic information into language model training. The improvements were not limited to isolated runs, but appeared consistently across ten random seeds and across all three model families. This makes it unlikely that the effect can be explained by favourable initialisation alone. More broadly, the findings show that behavioural similarity signals can shape embedding geometry in a predictable and measurable way, while also making clear that the scale and spread of these changes depend on the model architecture.

This experiment therefore represents an important step in the broader aims of the project. Earlier stages of the work focused on analysing the relationship between language model predictions and neural signals during comprehension. The RSR experiments extend that line of inquiry by exploring whether cognitively grounded similarity signals can actively shape

language model representations during training. In that sense, Experiment 3 marks a shift from comparison to intervention. Rather than simply asking whether model representations resemble human semantic organisation, it tests whether human semantic structure can be used to improve those representations directly.

A full mathematical specification of the RSR objective, along with additional implementation details and extended results, is provided in the accompanying draft paper, which is included in the appendix for reference.

4 Future Research Plan

4.1 Near-Term Plan: Representational Similarity Regularisation with Human Similarity Data

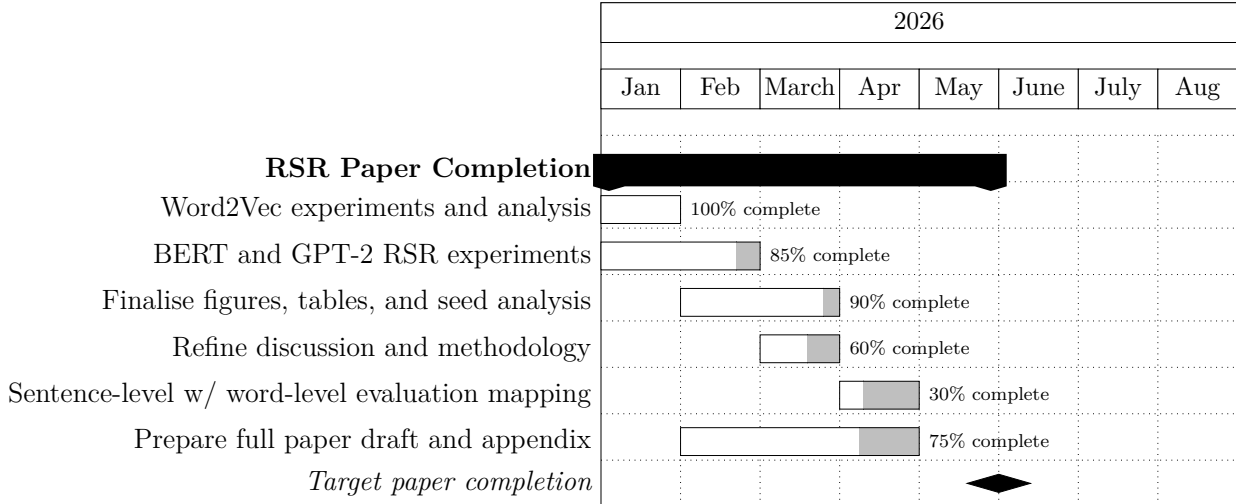
The most immediate next stage of the project is to complete and refine the current paper on Representational Similarity Regularisation (RSR). This work is already well advanced and currently provides the clearest proof of concept for the broader optimisation strand of the PhD. The main aim of the paper is to test whether behavioural similarity structure can be incorporated directly into language model training, and whether this leads to better alignment with human semantic judgements across different model architectures.

At present, the experimental framework has been implemented for Word2Vec, BERT, and GPT-2. Initial findings suggest that RSR produces consistent improvements on SimLex-999 across multiple random seeds, and that these gains are visible in both static and contextual models. For Word2Vec, the strongest improvements appear in parts of the embedding space that are more directly shaped by the similarity datasets. For BERT and GPT-2, the improvements seem to extend more broadly, which suggests that representational changes may spread through the pre-trained semantic manifold.

The remaining technical work on the paper is relatively focused. The main outstanding task is to refine the extraction procedure for the contextual models so that sentence-level representations can be more cleanly mapped to word-level representations for evaluation. While BERT and GPT-2 operate over sentence-level context, evaluation benchmarks such as SimLex-999 are defined over isolated word pairs. This creates a mismatch between training and evaluation. The aim is therefore not to move away from sentence-level representations, but to develop a principled method for extracting or aggregating token-level embeddings from contextual inputs in a way that allows fair comparison with word-level similarity judgements. This will make the evaluation more methodologically consistent with the Word2Vec setup and should strengthen the interpretation of the transformer results.

This near-term work matters for two main reasons. First, it delivers a complete experimental contribution in its own right, showing that similarity-based regularisation is a viable training-time mechanism for language models. Second, it provides a methodological base for the later stages of the PhD, where the source of similarity supervision can be extended beyond behavioural judgements to neural data derived from human brain recordings. In that sense, the current RSR paper forms a clear link between the earlier brain-model comparison experiments and the longer-term goal of cognitively grounded model optimisation.

4.2 Near-Term Timeline for the RSR Paper



4.3 Longer-Term Plan: Brain-Aligned Optimisation and Cross-Lingual Extension

Beyond the current RSR paper, the broader aim of the PhD is to investigate whether signals derived from human cognition can be used not only to analyse language models, but also to guide their optimisation more directly. The longer-term plan develops along three connected directions: higher-temporal-resolution neural comparison using MEG, cross-lingual investigation of semantic and neural alignment, and the incorporation of actual brain data into representational similarity regularisation.

The first of these directions involves the inclusion of MEG data. While the current fMRI experiments are useful for identifying broad spatial correspondences between model signals and brain activity, they are constrained by relatively low temporal resolution. MEG, by contrast, offers millisecond-level timing information, making it possible to track the unfolding of semantic and predictive processes in much finer detail. Incorporating MEG would allow the project to examine not only whether model representations align with brain responses, but also when this alignment occurs, and whether different model architectures correspond to different stages of neural processing. This would provide a much stronger form of cognitive grounding for the project.

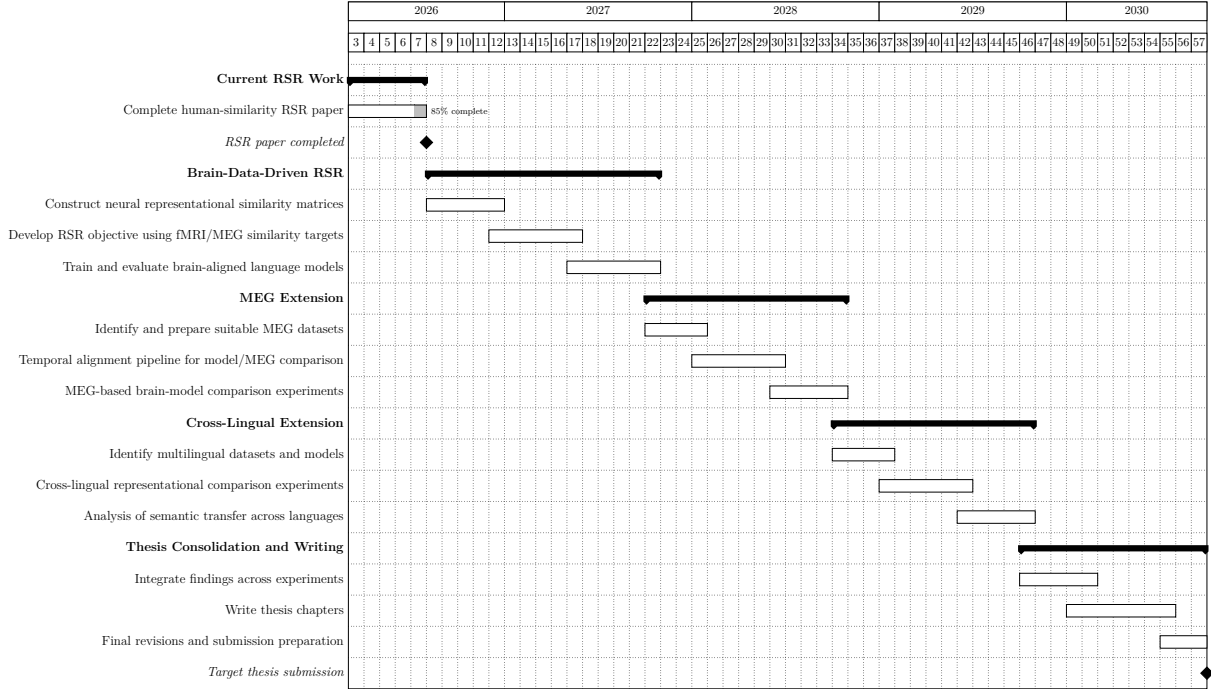
A second major direction is cross-lingual extension. Most of the work completed so far has focused on English, both in terms of model evaluation and the neural datasets available. An important open question, however, is whether the relationships observed between language models and human semantic processing generalise across languages with different syntactic and lexical structures. Extending the project in a cross-lingual direction would make it possible to test whether representational alignment is language-specific or whether it reflects deeper and more general properties of semantic organisation. This could involve comparing English and German data, using multilingual models, or examining whether similarity-based constraints learned in one language transfer to another. Such an extension would also help address broader questions of generalisability and bias in cognitively informed optimisation.

The third, and most ambitious, direction is to use actual human brain data as the source of representational similarity supervision. The current RSR experiments rely on behavioural similarity benchmarks such as MEN, SimVerb-3500, THINGS, and SimLex-999. These resources are valuable because they provide scalable and interpretable human judgements of semantic similarity. However, the wider motivation of the PhD is to understand whether neural representations of meaning can also be used to inform model optimisation. Prior work in brain decoding and encoding, including studies by Leila Wehbe [12], has demonstrated that distributed patterns of brain activity during naturalistic language comprehension contain rich and structured semantic information that can be systematically related to linguistic stimuli. Building on this line of work, a future version of RSR could be based on representational similarity matrices derived directly from fMRI or MEG recordings, for example from datasets such as Narratives. Instead of asking only whether model RDMs resemble brain RDMs after training, the objective would be to train models so that their internal geometry becomes more consistent with the similarity structure observed in neural data. Concretely, this would involve constructing brain-derived similarity targets across stimuli and incorporating them into the optimisation process in a similar way to behavioural similarity signals, but with the additional constraint that the supervision reflects population-level neural organisation rather than explicit human judgements.

This would represent a substantial shift from comparison to intervention, in which brain-derived signals move from being a passive evaluation metric to an active training signal. Rather than treating brain alignment purely as an evaluative measure, it would become part of the optimisation process itself. A framework of this kind could encourage models to develop representations that are more human-like not only in behavioural terms, but also in the structure of their internal activations. This is a challenging direction, both technically and conceptually, but it is also one of the most promising aspects of the longer-term project.

Taken together, these longer-term directions define the broader trajectory of the PhD. The current work has already established the feasibility of comparing model predictions with neural activity and of using behavioural similarity to regularise training. The next phase will ask whether richer neural signals, broader linguistic coverage, and brain-derived similarity structure can be brought together within the same framework to produce language models that are more cognitively grounded, more interpretable, and more closely aligned with human semantic processing.

4.4 Long-Term PhD Timeline



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A Behavioural Similarity Datasets

This section provides additional detail on the behavioural datasets used for representational similarity regularisation experiments.

A.1 MEN Dataset

The MEN dataset [17] contains approximately 3,000 word pairs with human similarity judgements. It was designed to evaluate general semantic relatedness and includes a wide range of common concepts. MEN is particularly useful for capturing broad associative relationships, although it does not strictly distinguish between similarity and relatedness.

A.2 SimLex-999

SimLex-999 [20] is a widely used benchmark dataset for evaluating semantic representations. It consists of 999 word pairs annotated with similarity ratings provided by human participants. The dataset was constructed to cover a range of parts of speech, including nouns, verbs, and adjectives, and includes pairs spanning different levels of concreteness and frequency. Human ratings were collected under controlled conditions and subsequently normalised to produce a consistent similarity scale. SimLex-999 is commonly used to assess how well computational models capture human judgements of similarity by measuring the correlation between model predictions and the annotated scores.

A.3 SimVerb-3500

SimVerb-3500 [18] extends similarity evaluation to the verbal domain, containing 3,500 verb pairs annotated with human similarity scores. It addresses an important gap in earlier datasets, which focused primarily on nouns, and enables more comprehensive evaluation of model representations across parts of speech.

A.4 THINGS Dataset

The THINGS dataset [19] is substantially larger and more diverse than traditional similarity benchmarks. It contains 1,854 object concepts and over 26,000 naturalistic images, along with behavioural similarity judgements. Importantly, THINGS is inherently multimodal, linking linguistic concepts to visual representations.

In addition to behavioural data, THINGS has been extended in related work to include neural recordings such as fMRI responses to object stimuli. This makes it particularly valuable for future work aiming to align language model representations with both behavioural and neural similarity structure. Its scale and multimodal nature therefore make it a promising resource for cognitively grounded model optimisation.

B Representational Similarity Regularisation Paper

This appendix contains the full draft paper on Representational Similarity Regularisation (RSR), which forms the primary experimental contribution of this report. The paper provides a complete description of the methodology, mathematical formulation of the objective function, experimental setup, and extended results across Word2Vec, BERT, and GPT-2 models.

While the main body of this report summarises the key ideas and findings, the appendix is included to provide full technical detail and to demonstrate the scope of the work in a format suitable for future publication.

Representational Similarity Regularisation During Language Model Training: A Proof-of-Concept Study

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March 2025

Abstract

Representational Similarity Regularisation (RSR) introduces behavioural similarity structure directly into model learning. While similarity analysis has been used to compare models and cognitive data [1], its integration into optimising models remains underexplored. This paper provides a proof-of-concept demonstration that RSR can be incorporated into language model training. We evaluate RSR across static (Word2Vec [2]) and contextual (BERT [3], GPT-2 [4]) models and observe consistent improvements in alignment with human similarity judgements. For the transformer experiments, BERT and GPT-2 are used in a controlled word-level setting in which isolated lexical items are encoded by pretrained contextual backbones and then fine-tuned with an RSR objective. Notably, these contextual models still exhibit generalisation beyond directly supervised vocabulary, whereas improvements in static embeddings remain localised. These findings establish RSR as a viable training-time regulariser and suggest that pre-existing representational structure plays a critical role in how similarity constraints propagate through embedding spaces.

1 Introduction

Language models learn semantic structure from patterns of co-occurrence [2]. Although these representations are effective for many downstream tasks, their similarity structure does not always align with human judgements of semantic similarity. Benchmarks such as SimLex-999 [5] reveal systematic mismatches between distributional relatedness and genuine conceptual similarity.

Representational Similarity Regularisation (RSR), originally proposed by Ormerod [6], offers a way to address this mismatch by introducing an additional learning signal derived from behavioural similarity data. Rather than analysing similarity structure after a model has been trained using Representational Similarity Analysis (RSA) [1], RSR encourages the embedding space to reflect human similarity directly during training.

The central question of this paper is straightforward: can representational similarity be incorporated into language model training in a stable and effective way? A secondary methodological question follows from this for the transformer experiments: if BERT and GPT-2 are typically trained to model sequences, can they still provide useful word-level semantic representations in a controlled lexical setting, and can those representations be reshaped through similarity-based fine-tuning?

We provide a proof-of-concept answer across three architectures:

- Word2Vec trained from scratch using interleaved similarity updates [2],
- BERT fine-tuned with a similarity objective [3],
- GPT-2 fine-tuned with the same objective [4].

2 Related Work

Representational Similarity Analysis (RSA), introduced by Kriegeskorte et al. [1], compares similarity structures across neural, behavioural, and computational representations. RSA has become standard in cognitive neuroscience and computational modelling.

Similarity-based objectives have more recently been incorporated into optimisation. Ormerod [6] demonstrated that representations can be regularised to align with neural similarity structure during fine-tuning. That work established the base RSR framework adopted in this paper. The present study differs in three respects: (i) the focus is word-level similarity (rather than sentence-level), (ii) the supervision signal is behavioural rather than neural, and (iii) we evaluate across both static and contextual architectures.

Word2Vec introduced the skip-gram objective for learning static embeddings from co-occurrence statistics [2]. BERT and GPT-2 extended distributional learning to contextual transformer architectures via masked and autoregressive language modelling [3, 4]. The question addressed here is whether similarity-alignment objectives can be applied consistently across these architectures.

3 Data and Models

Human Similarity Data: Similarity supervision is drawn from a combination of the datasets titled 'MEN' [7], 'SimVerb-3500' [8], and 'THINGS' [9]. Each dataset is normalised independently, filtered to retain only word pairs that can be represented by the relevant model or tokenizer, and converted into a unified pool of similarity-labelled pairs. During training, mini-batches of pairs are sampled from this combined pool, ensuring that supervision spans many different semantic domains.

Architectures: Three model classes are evaluated:

- Skip-gram Word2Vec (static embeddings),
- BERT (bidirectional transformer),

- GPT-2 (autoregressive transformer).

The RSR objective is architecture-agnostic and operates on the similarity structure of learned representations.

Why a Word-Level Setting? The behavioural datasets used for supervision and evaluation consist of lexical pairs rather than full sentences. For that reason, the experiments were designed around word-level similarity rather than contextual sentence understanding. This choice makes the comparison across architectures more controlled: Word2Vec, BERT, and GPT-2 are all asked to produce a representation for the same lexical item, and the question is whether RSR can move that representation closer to human semantic judgements. Although BERT and GPT-2 are normally associated with sentence processing, they still contain rich lexical and subword structure in their hidden states. The present experiments therefore do not ask whether the models perform full contextual comprehension in isolation, but whether their pretrained representational machinery can be used as a basis for improving lexical semantic geometry.

4 Representational Similarity Regularisation

Representational Similarity Regularisation (RSR), originally proposed by Ormerod [6], introduces behavioural similarity structure directly into model optimisation. Rather than analysing similarity structure after model training using Representational Similarity Analysis [1], RSR incorporates similarity alignment as an explicit training objective. The formulation is architecture-agnostic and can be applied to static or contextual models.

4.1 Similarity Prediction

RSR operates on pairs of items for which human similarity judgements are available. For a mini-batch of N labelled pairs (i, j, s_{ij}) , the model produces vector representations $\mathbf{v}_i$ and $\mathbf{v}_j$ for each word under the current parameters.

Model similarity is computed as the cosine similarity between the corresponding embedding vectors. Cosine similarity measures the angular alignment between vectors and is invariant to vector norm, making it a natural proxy for semantic proximity in distributional embedding spaces.

For a batch, this yields a vector of predicted similarities $\hat{\mathbf{s}} \in \mathbb{R}^N$, which can be compared directly to the vector of human similarity judgements $\mathbf{s} \in \mathbb{R}^N$. RSR operates on these two vectors to align the geometry of the embedding space with behavioural similarity structure.

4.2 Rank-Based Similarity Loss

Rather than matching raw similarity numbers, RSR aligns rank structure. Following Ormerod [6], we optimise a differentiable approximation to Spearman rank correlation:

$$L_{\text{rsr}} = 1 - \rho_{\text{soft}}(\hat{\mathbf{s}}, \mathbf{s}), \quad (1)$$

where ρ_{soft} computes Pearson correlation between differentiable (soft) rank transforms of $\hat{\mathbf{s}}$ and $\mathbf{s}$.

This objective focuses directly on relative semantic structure and avoids scale sensitivity across different datasets.

4.3 Integration with Training

RSR augments a model’s representational learning with an additional similarity-alignment term. Conceptually, a combined objective can be written as

$$L = L_{\text{task}} + L_{\text{rsr}}, \quad (2)$$

where L_{task} denotes the model’s original training loss (e.g., skip-gram for Word2Vec, masked language modelling for BERT, or autoregressive language modelling for GPT-2), and L_{rsr} is the rank-based similarity regularisation term defined above.

This formulation makes RSR architecture-agnostic: it operates purely on the geometry of learned representations and can therefore be integrated into any model that produces vector embeddings for items. In the present study, however, the practical integration differs across architectures. For Word2Vec, similarity updates are interleaved with the standard skip-gram objective during training from scratch. For BERT and GPT-2, the models begin from pretrained parameters and are then fine-tuned using the RSR objective over a mostly frozen backbone. Thus, Equation 3 should be understood as the general conceptual form of the method rather than a literal statement that every experiment jointly optimises the original language-modelling loss at every step.

5 Word2Vec from Scratch

We first evaluate RSR in the simplest setting: static word embeddings trained from scratch.

5.1 Distributional Objective

We use the standard skip-gram objective with negative sampling [2]. The skip-gram model learns static word embeddings by predicting surrounding context words given a central target word within a fixed window, thereby encoding distributional co-occurrence structure.

For a target word i and observed context word j , the loss is:

$$L_{\text{dist}} = -\log \sigma(\mathbf{v}_i^\top \mathbf{u}_j) - \sum_{k=1}^K \log \sigma(-\mathbf{v}_i^\top \mathbf{u}_{n_k}), \quad (3)$$

where $\mathbf{v}_i$ and $\mathbf{u}_j$ are target and context embeddings, K negative context words n_k are sampled from a smoothed unigram distribution, and $\sigma(\cdot)$ denotes the logistic sigmoid. Frequent words are subsampled following standard practice.

5.2 Evaluation Steps

We evaluate the resulting W2V models using SimLex-999 [5], a benchmark specifically designed to measure genuine semantic similarity rather than general relatedness. SimLex-999 provides human similarity ratings for 999 word pairs spanning nouns, verbs, and adjectives.

Performance is measured using Spearman rank correlation between model cosine similarities and human similarity scores. Spearman correlation is appropriate because both SimLex-999 and the RSR objective emphasise relative ordering of similarities rather than absolute magnitudes.

Importantly, SimLex-999 is not used during RSR training. It therefore serves as a held-out evaluation of how well similarity structure generalises beyond the supervision pool. In the Word2Vec analysis, SimLex pairs are also partitioned according to their relationship to the THINGS concept set: pairs for which both words were directly covered by THINGS supervision, pairs for which only one word overlapped, and pairs for which neither word belonged to the supervised concept pool. This partitioning is helpful because it reveals whether gains are confined to directly supervised concepts or whether they spread more broadly through the learned embedding space.

5.3 Results

Table 1 reports SimLex-999 [5] performance averaged across 10 random seeds using interleaving frequency $k = 10$.

Table 1: Word2Vec SimLex-999 results (10 seeds, $k = 10$).

Category	Vanilla	RSR	Δ
All pairs	0.273	0.413	+0.140
Both in RSR	0.148	0.517	+0.369
One in RSR	0.409	0.404	-0.005
Neither in RSR	0.299	0.299	+0.000

RSR produces a substantial overall improvement (+0.141). The effect is concentrated in pairs where both words were included in the RSR similarity pool (+0.368). Pairs containing one or no supervised words show little change.

This indicates that, when training embeddings from scratch, similarity supervision primarily reshapes the directly supervised region of the embedding space rather than propagating globally.

To assess robustness, Figure 1 plots vanilla versus RSR performance for each of the 10 seeds.

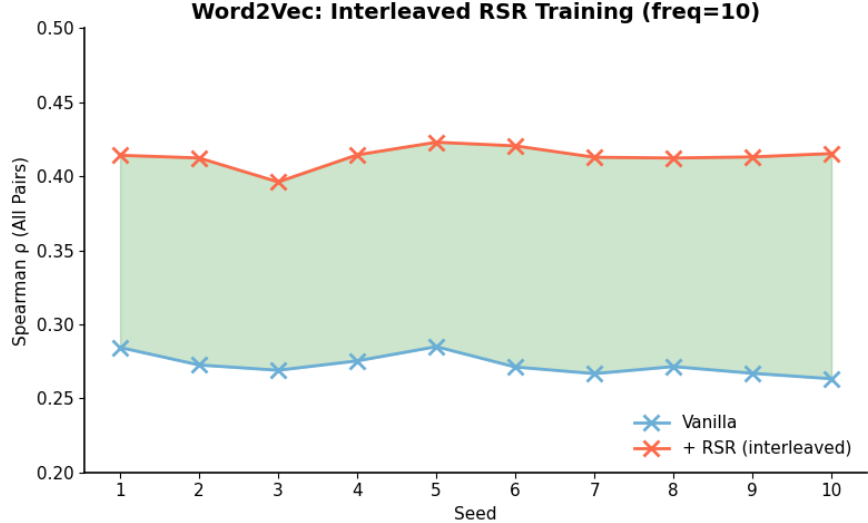


Figure 1: Vanilla vs. RSR SimLex-999 performance across 10 seeds for Word2Vec ($k = 10$). Each point represents one seed. All seeds show improvement under RSR.

All points lie above the diagonal, indicating that RSR improves performance consistently across random initialisations.

6 Contextual Transformers

6.1 Setup

For both BERT and GPT-2, a lightweight word-embedding wrapper was constructed around the pretrained transformer backbone. In each case, the hidden dimensionality of the pretrained model is 768, and a learned linear projection head maps this representation into a 128-dimensional space used for similarity comparison. This projection step serves two purposes. First, it provides a compact semantic space in which cosine similarity can be measured efficiently across many word pairs. Second, it limits the amount of trainable capacity introduced during fine-tuning, which is desirable in a proof-of-concept setting where the goal is to test whether RSR can steer an existing representation rather than to retrain the entire model.

Most of the pretrained transformer is frozen. In BERT, the embedding layer, the pooler, and the first 11 encoder layers are frozen, leaving only the final encoder block and the projection head trainable. In GPT-2, the token and positional embeddings, the first 11 transformer blocks, and the final layer normalisation are frozen, again leaving only the final transformer block and projection head trainable. This design isolates the effect of similarity-based fine-tuning: if changes are observed after training, they cannot be attributed to wholesale re-training of the underlying language model, but instead to relatively targeted adjustment of the top-level representational geometry.

Word representations are extracted in a deliberately simple and consistent way. Each lexical item is tokenised on its own. For BERT, special tokens are added and the hidden states

corresponding to the word’s subword pieces are mean-pooled, excluding the surrounding special tokens. For GPT-2, a leading space is added before tokenisation so that the resulting byte-pair encoding more closely reflects normal word boundaries, and the hidden states of the resulting subword pieces are again mean-pooled. Mean-pooling is not the only possible choice, but it has two practical advantages here: it provides a stable fixed-size representation for words that may split into different numbers of subword units, and it avoids introducing additional architectural assumptions such as selecting a special classification token or designing a more complex composition module.

RSR is then applied during fine-tuning using the cosine similarity and soft Spearman objective defined in Section 4. Importantly, in these transformer experiments the optimisation is driven by the similarity objective alone rather than jointly with masked language modelling or autoregressive next-token prediction. The models therefore begin from pretrained linguistic structure and are then selectively reshaped by behavioural similarity supervision.

6.2 Training Procedure

Each experiment is repeated across 10 seeds. For every seed, the vanilla pretrained model is first evaluated on SimLex-999 to establish a baseline. The model is then fine-tuned for 200 RSR epochs using an Adam optimiser with learning rate 10^{-3} . At each epoch, up to 10,000 labelled word pairs are sampled from the combined MEN, SimVerb-3500, and THINGS pool. Unique words from that sample are embedded in batches, cosine similarities are computed for each labelled pair, and the model is updated to maximise soft Spearman agreement between predicted and human similarity rankings.

This training procedure reflects a compromise between feasibility and interpretability. Sampling a large but bounded set of pairs at each epoch keeps computation manageable while still exposing the model to diverse semantic relations. Repeating the entire procedure across seeds is important because the projection head is randomly initialised and the exact sample drawn from the supervision pool varies across epochs. If improvements were only seen for one or two seeds, the results could plausibly be attributed to favourable initialisation. Consistent gains across seeds, by contrast, provide evidence that the effect is systematic.

6.3 Results

Table 2 presents SimLex-999 performance for BERT and GPT-2, averaged across 10 seeds.

Table 2: BERT and GPT-2 SimLex-999 results after RSR fine-tuning (10 seeds).

Category	BERT			GPT-2		
	Vanilla	RSR	Δ	Vanilla	RSR	Δ
All pairs	0.148	0.281	+0.133	0.097	0.226	+0.129
Both in RSR	0.021	0.296	+0.275	0.025	0.246	+0.221
One in RSR	0.240	0.321	+0.081	0.129	0.251	+0.122
Neither in RSR	0.178	0.355	+0.177	0.129	0.295	+0.166

Both contextual models show clear improvement under RSR. Unlike Word2Vec, gains

extend beyond directly supervised vocabulary. In particular, performance improves substantially for pairs where neither word was included in the similarity training pool (+0.177 for BERT, +0.166 for GPT-2). This is a particularly important, as it suggests that the transformer-based representations are not merely memorising the supervised lexical pairs. Instead, the similarity constraint appears to reshape a broader region of the representational space in a way that benefits held-out lexical relationships as well.

Figure 2 visualises performance across seeds.

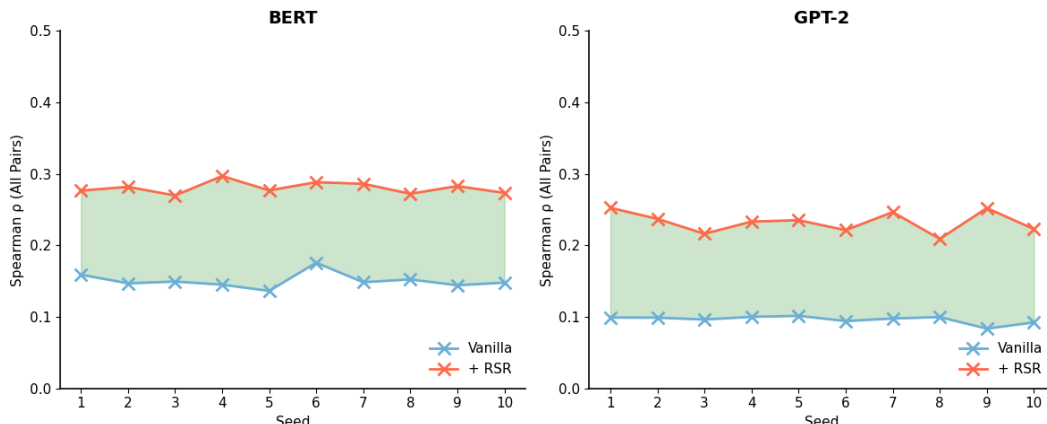


Figure 2: Vanilla vs. RSR SimLex-999 performance across 10 seeds for BERT and GPT-2. Improvements are consistent across random initialisations for both architectures.

All points lie above the diagonal, indicating that RSR reliably reshapes similarity structure in contextual models.

The transformer results should also be interpreted in light of the word-level design choice. Even though BERT and GPT-2 are normally used in sentence-level settings, the present experiment shows that their pretrained hidden-state geometry already contains lexical information rich enough to benefit from behavioural similarity regularisation. This does not mean that the models cease to be contextual or that context is unimportant; rather, it means that a contextual architecture can still be used as a powerful word encoder when the task itself is lexical. In that sense, the experiment isolates one particular aspect of these models - their capacity to represent word meaning-and demonstrates that this aspect can be improved in a controlled, cognitively motivated way.

7 Discussion

RSR improves similarity alignment across all three architectures. However, the manner in which improvements propagate differs markedly.

For Word2Vec trained from scratch, gains are concentrated on directly supervised vocabulary. Embeddings for unseen words remain largely unchanged. In contrast, BERT and GPT-2 exhibit substantial improvements even for word pairs that were never included in the similarity training pool.

This contrast suggests that architecture influences how similarity constraints propagate through representational space. Pre-trained transformers encode dense, globally organised semantic manifolds learned from large corpora prior to any RSR fine-tuning. When similarity structure is adjusted locally, those adjustments can diffuse through pre-existing lexical and subword-sensitive organisation already present in the model. Static embeddings trained from scratch lack such global structure, and similarity constraints remain more localised.

RSR therefore appears architecture-agnostic in formulation, but architecture-dependent in how its effects spread.

8 Implications and Future Directions

These results suggest that behavioural similarity signals can do more than improve a narrow benchmark. They appear capable of reshaping representational geometry in a way that is both structured and, at least in the transformer experiments, capable of generalising beyond the exact words seen during training. This is especially interesting because it implies that the supervision signal is not simply being memorised. Instead, the models seem to be adjusting broader semantic relationships in response to relatively local constraints.

That observation matters because it points to a practical advantage of RSR. If targeted human similarity judgements can produce wider changes in representational space, then behavioural supervision may not need to be exhaustive in order to be useful. A relatively modest amount of carefully chosen similarity data may be enough to nudge a model toward more human-like semantic organisation, particularly when the model already contains rich structure from pre-training.

This raises the possibility of using RSR as a lightweight way of steering language models in settings where semantic precision matters. Examples include lexical semantics research, cognitive modelling, and educational or clinical language technologies, where small differences in meaning can have disproportionate importance. More broadly, RSR offers a simple way of incorporating structured human knowledge into neural models without requiring a new architecture.

There are several natural directions for future work. One is to test whether the representational improvements observed here translate into gains on downstream language tasks rather than remaining confined to similarity benchmarks. Another is to investigate whether RSR can be incorporated earlier in training, including during larger-scale pre-training rather than only at the fine-tuning stage. A further question is theoretical: if similarity constraints spread more broadly in contextual transformers than in static embeddings, what does that reveal about the internal geometry of those models and the way semantic information is organised within them?

9 Conclusion

This work demonstrates that representational similarity regularisation can be incorporated into language model training and produces reliable improvements in human similarity alignment. Gains are consistent across seeds and architectures. Static embeddings benefit directly

from supervised alignment, while contextual models exhibit broader generalisation effects. RSR offers a simple and flexible mechanism for integrating behavioural similarity signals into modern language models.

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